E-Commerce Sales Data Transformation and Modeling using Power BI

Project Overview

This project focuses on preparing and modeling an E-commerce sales dataset using **Microsoft Power BI** and **Power Query Editor**. The objective was to transform raw data into a clean, structured, and analysis-ready data model by performing data cleansing, transformation, aggregation, and relationship modeling. The final dataset provides a reliable foundation for creating interactive dashboards and generating meaningful business insights.

Objectives

The primary objectives of this project were to:

- Import multiple datasets into Power BI.
- Perform data cleaning and transformation using Power Query.
- Standardize data formats and improve data quality.
- Create calculated and conditional columns.
- Merge related datasets using appropriate joins.
- Handle missing values and duplicate records.
- Aggregate data to generate business summaries.
- Build relationships between tables to support efficient data analysis.
- Prepare an optimized data model for dashboard development.

Dataset Description

The project uses three related datasets.

1. List of Orders

Contains customer and order-level information.

Columns

- Order ID
- Order Date
- Customer Name
- City
- State

2. Order Details

Contains detailed product-level information for each order.

Columns

- Order ID
- Category
- Sub-Category
- Quantity
- Amount
- Profit

3. Sales Target

Contains monthly sales targets by product category.

Columns

- Category
- Month of Order Date
- Target

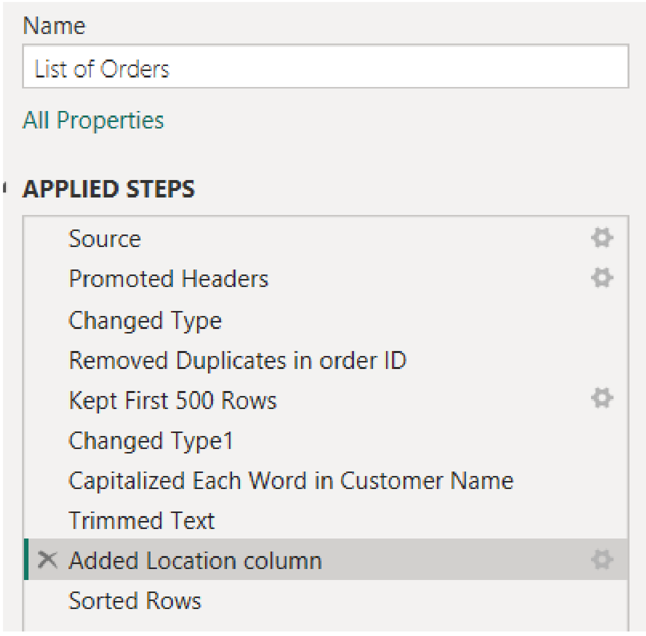
Data Import

All three CSV files were imported into Power BI Desktop and opened in **Power Query Editor** using the **Transform Data** option for preprocessing.

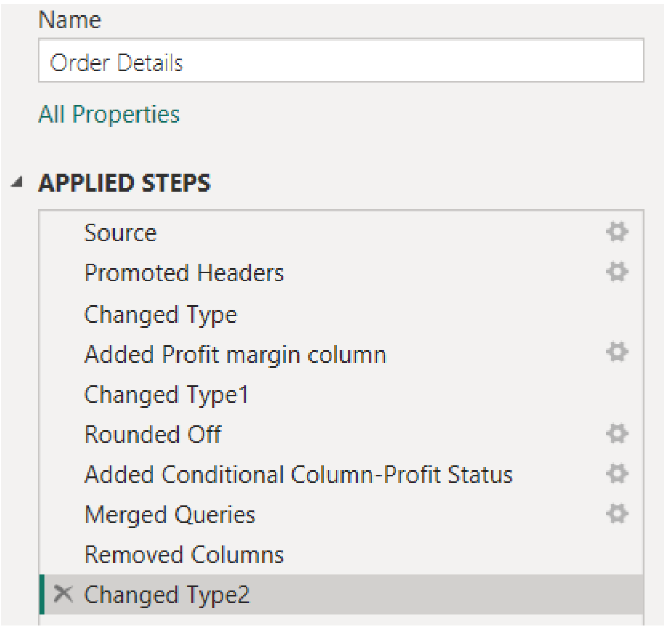
Data Cleaning and Transformation

The following data cleaning and Transformation steps were performed to improve data quality and consistency.

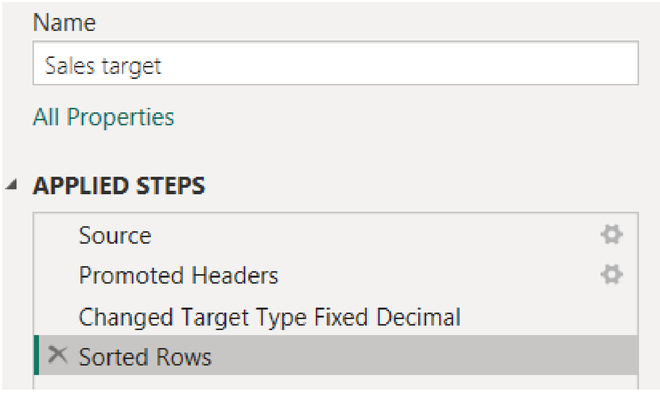
1. List of Orders Table



2. Order Details Table

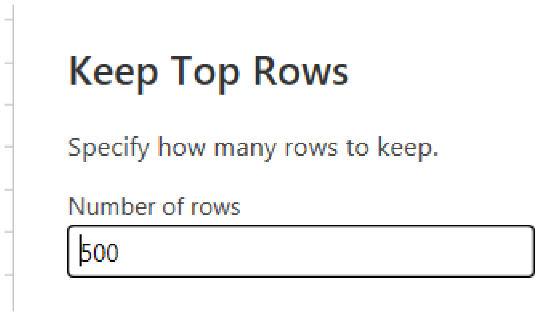


3. Sales Target Table



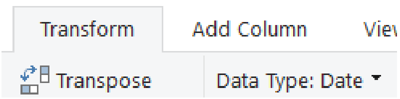
Row Limiting

- Restricted the **List of Orders** dataset to the first **500 records** as required.

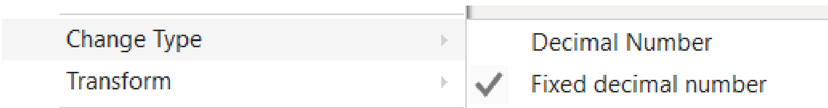


Data Type Conversion

- Converted **Order Date** to the **Date** data type.

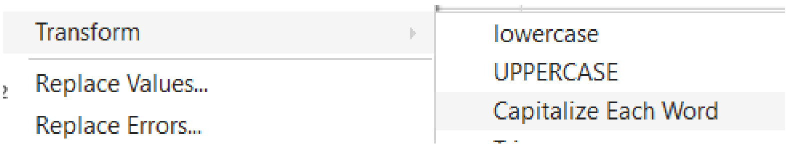


- Converted **Amount** and **Target** columns to **Fixed Decimal Number**.



Text Standardization

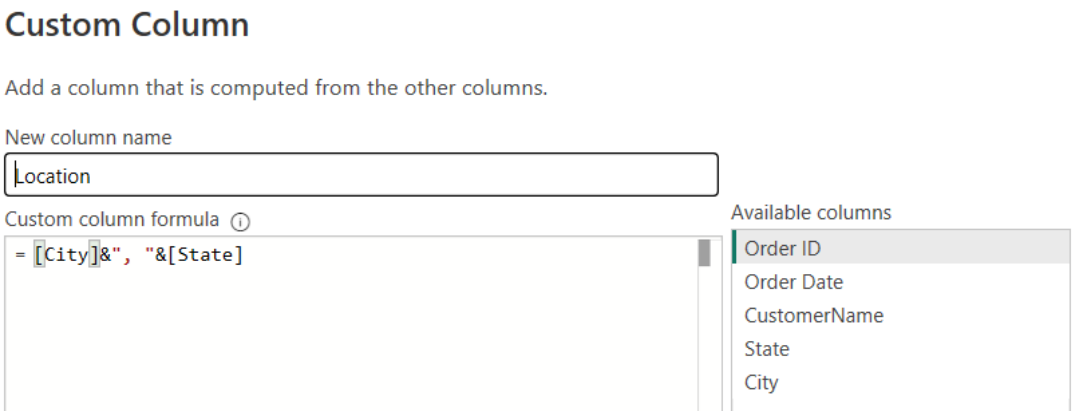
- Formatted the **Customer Name** column to **Proper Case** to ensure consistent capitalization.



Feature Engineering

Location Column

Created a new column named **Location** by combining the **City** and **State** columns.

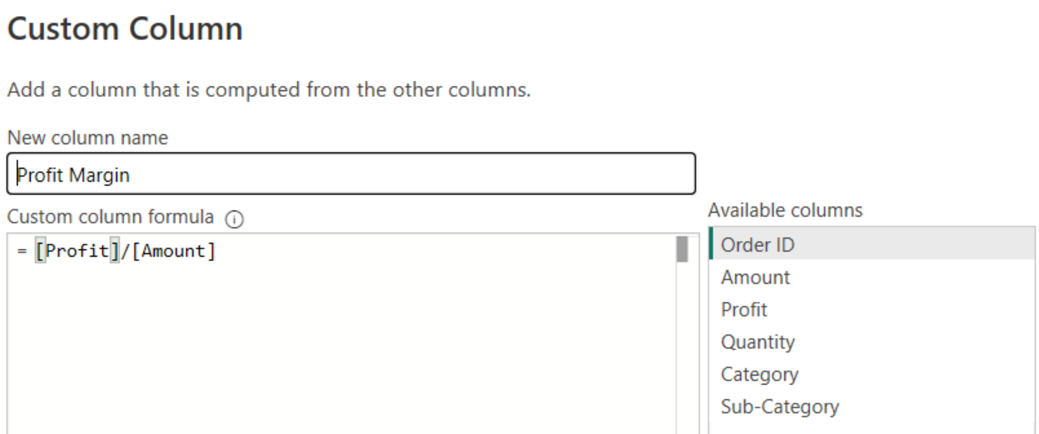


Profit Margin

Created a custom column to calculate the profit margin using the formula:

Profit Margin = Profit / Amount

The result was formatted as a **percentage** for better readability.



Profit Status

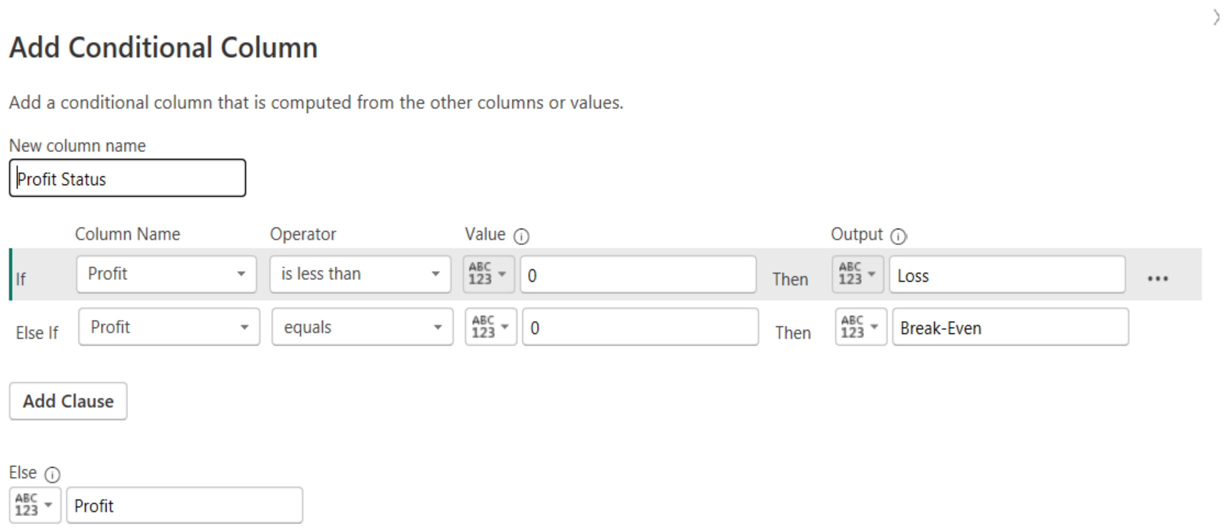
Created a conditional column named **Profit Status** using the following business rules:

Condition Status

Profit < 0 Loss

Profit = 0 Break-Even

Profit > 0 Profit



Data Integration

The **List of Orders** and **Order Details** tables were merged using **Order ID**.

- Join Type: **Left Outer Join**
- Output Table: **Orders Data**

The merged dataset combines customer information with detailed product transactions, creating a comprehensive sales dataset.

Merge

Select tables and matching columns to create a merged table.

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	*1,275.00*	*-1148*	*7*	Furniture	Bookcases	*-90.00%*	Loss
B-25601	*66.00*	*-12*	*5*	Clothing	Stole	*-18.00%*	Loss
B-25601	*8.00*	*-2*	*3*	Clothing	Hankerchief	*-25.00%*	Loss
B-25601	*80.00*	*-56*	*4*	Electronics	Electronic Games	*-70.00%*	Loss
B-25602	*168.00*	*-111*	*2*	Electronics	Phones	*-66.00%*	Loss

List of Orders

Order ID	Order Date	CustomerName	State	City	Location
B-25601	*01-04-2018*	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
B-25602	*01-04-2018*	Pearl	Maharashtra	Pune	Pune, Maharashtra
B-25603	*03-04-2018*	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
B-25604	*03-04-2018*	Divsha	Rajasthan	Jaipur	Jaipur, Rajasthan
B-25605	*05-04-2018*	Kasheen	West Bengal	Kolkata	Kolkata, West Bengal

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Data Quality Management

Missing Values

The datasets were examined for missing values.

The following strategies were applied based on business relevance:

- Replaced missing values where appropriate.
- Removed records containing critical missing information.
- Retained null values only when they did not affect the overall analysis.

Duplicate Records

Duplicate rows were identified and evaluated.

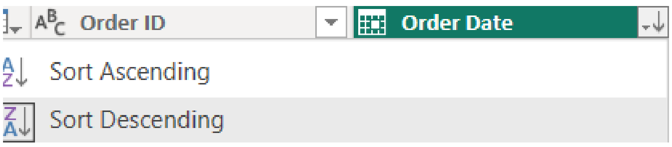
Handling strategy:

- Removed exact duplicate records.
- Retained repeated rows representing valid business transactions, such as multiple products within the same order.

Data Sorting and Filtering

To support business analysis:

- Orders were sorted by **Order Date** in descending order.



- Filters were applied to analyze sales for specific states, including **Tamil Nadu**.

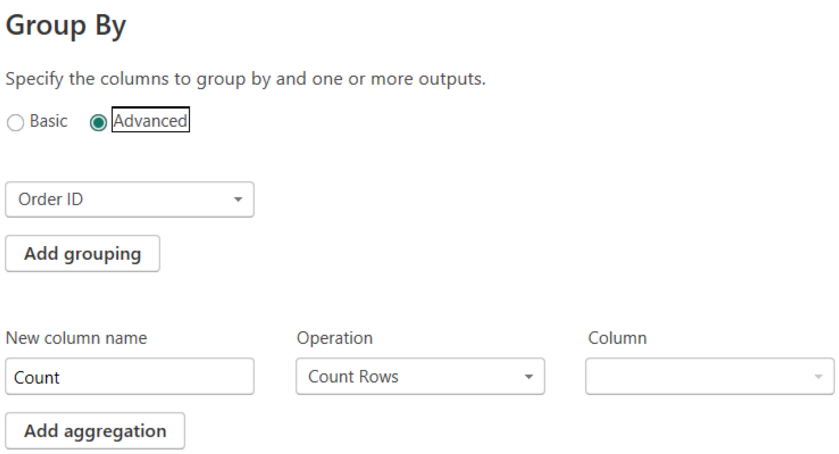
Data Aggregation

Order Details Summary

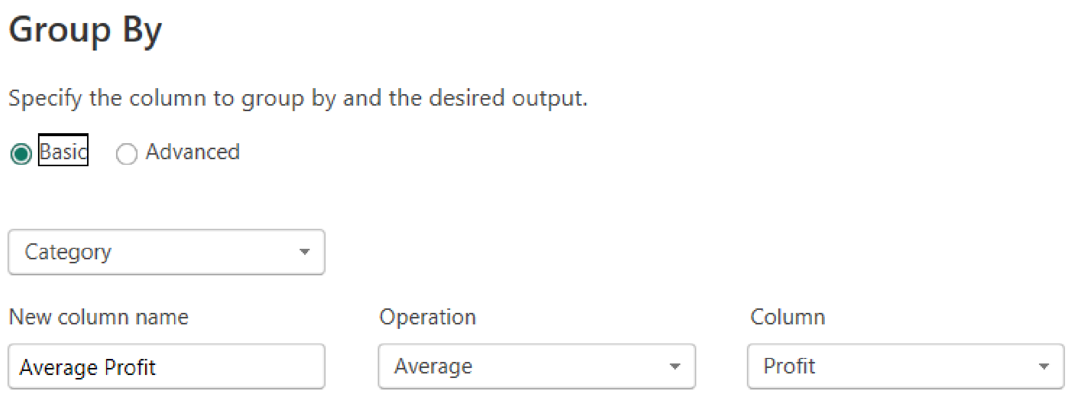
A duplicate of the **Order Details** table was created to generate summary statistics.

The following aggregations were performed:

Count of Order ID



Average Profit by Category



Output

	Category	Average Profit
1	Furniture	9.46
2	Clothing	11.76
3	Electronics	34.07

Total Amount by Sub-Category

Group By

Specify the column to group by and the desired output.

(●) Basic ( ) Advanced

Sub-Category

New column name	Operation	Column
Total Amount	Sum	Amount

Output

	Sub-Category	Total Amount
1	Bookcases	56,861.00
2	Stole	18,546.00
3	Hankerchief	14,608.00
4	Electronic Games	39,168.00
5	Phones	46,119.00
6	Saree	53,511.00
7	Trousers	30,039.00
8	Chairs	34,222.00
9	Kurti	3,361.00
10	T-shirt	7,382.00
11	Shirt	7,555.00
12	Leggings	2,106.00
13	Tables	22,614.00
14	Printers	58,252.00
15	Accessories	21,728.00
16	Furnishings	13,484.00
17	Skirt	1,946.00

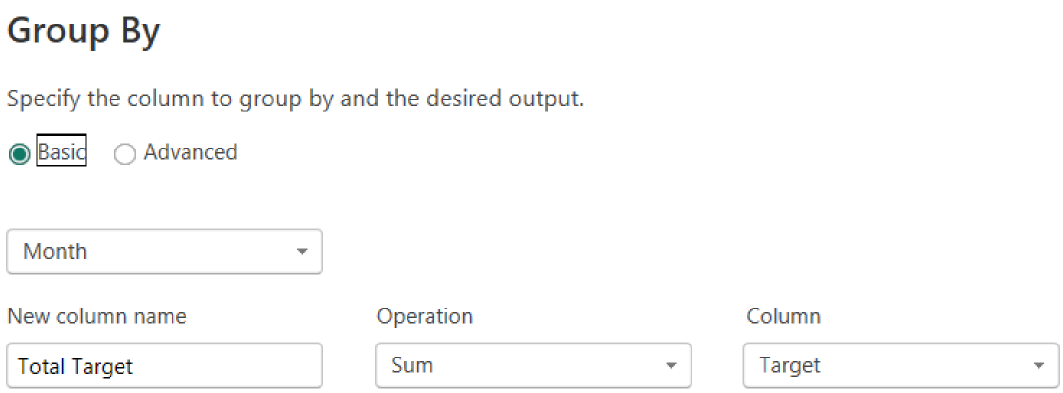
These summaries provide insights into order volume, profitability, and sales performance across product categories.

Sales Target Summary

A duplicate of the **Sales Target** table was created and grouped by **Month of Order Date**.

Aggregation performed:

Total Target (Sum)



Output

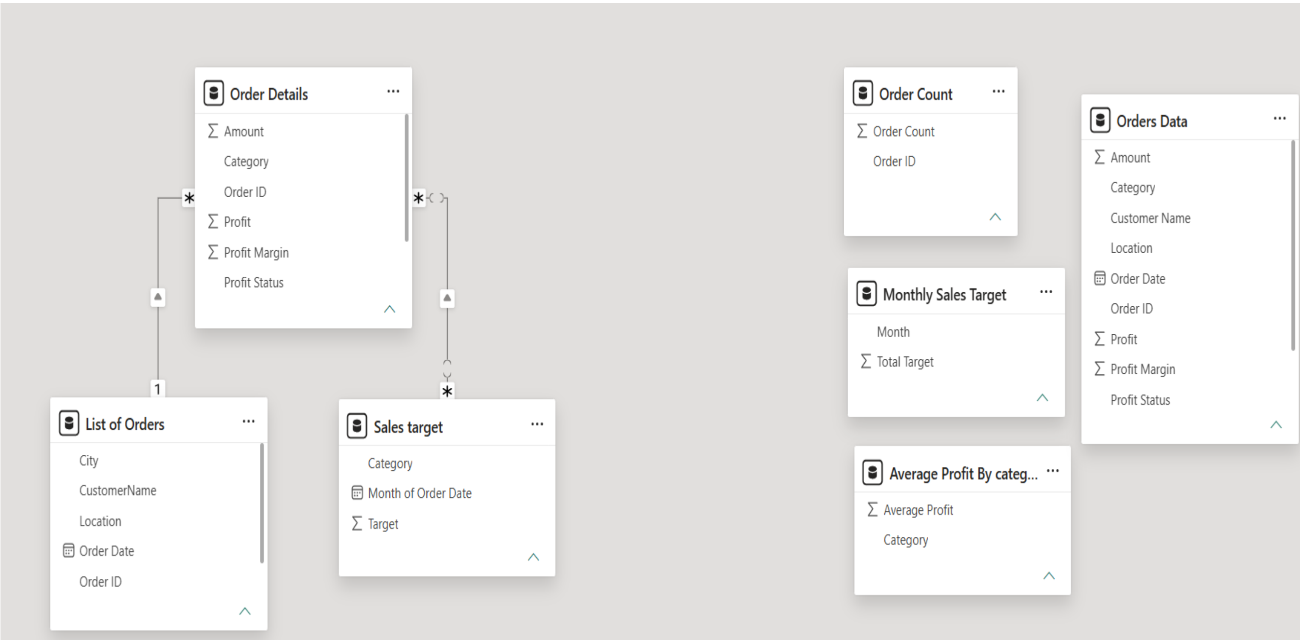
	Month	Total Target
1	April	31400
2	May	31500
3	June	31600
4	July	33800
5	August	33900
6	September	34000
7	October	36100
8	November	36300
9	December	36400
10	January	43500
11	February	43600
12	March	43800

This summary supports monthly sales target analysis and performance tracking.

Data Modeling

Relationships were established using the **Manage Relationships** feature in Power BI.

From Table	To Table	Join Column	Cardinality
List of Orders	Order Details	Order ID	One-to-Many (1:*)
Order Details	Sales Target	Category	Many-to-Many (:)



All relationships were configured as **Active** with **Single Cross Filter Direction** to ensure accurate data filtering and analysis.

Business Value

The prepared data model enables stakeholders to:

- Monitor sales performance efficiently.
- Compare actual sales with monthly sales targets.
- Analyze profit across categories and sub-categories.
- Evaluate customer purchasing trends.
- Perform regional sales analysis.
- Build interactive dashboards for informed business decision-making.

Tools and Technologies

- Microsoft Power BI Desktop
- Power Query Editor
- Data Modeling
- CSV Data Sources

Conclusion

This project demonstrates a complete data preparation and modeling workflow using Power BI. Through data cleansing, transformation, aggregation, and relationship modeling, raw E-commerce datasets were converted into a structured and analysis-ready data model. The final solution improves data quality, supports accurate reporting, and provides a strong foundation for creating interactive business intelligence dashboards and generating actionable insights.